

TM Forum Component

Product Usage Management

TMFC040

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Notice

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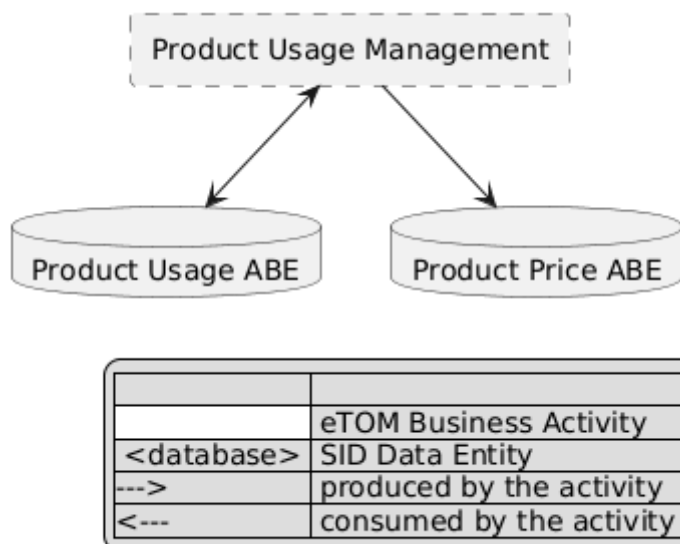
TMFC040 – Product Usage Management

1. Overview

Component Name	ID	Description	ODA Function Block
Product Usage Management	TMFC040	Product Usage Management component provides a standardized mechanism for creation, update, retrieval, import and export of a collection of usages. It allows a service provider to retrieve usage generated by a partner service platform in order to rate it, and provides rated usage to a partner for consumption follow-up purposes.	Core Commerce

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.2.16	L2	Product Usage Management	The Product Usage management processes encompass the functions required to guide, distribute, mediate, summarize, accumulate, and analyze Product Usage records. These processes may occur in real-time, near real-time (i.e. just at the end of the usage), or may be executed on a periodic basis. // Based on Service Usage, this process aims at identifying Product Usage. For example, for a Video on Demand where you can watch a video as many times as you want during 72 hours, several Service Usages might have been tracked (each time the user watches the video) and only one Product Usage will be identified for all Service Usages in the 72 hours after the first watch. // The guiding processes ensure that the Product Usage records used in the billing processes are appropriately related to the correct customer billing account and products. // The Product Usage records are edited and if necessary reformatted (mediated) to meet the needs of subsequent processes. The billing event records may also be enriched with additional data during this process.
1.2.16.1	L3	Product Usages	The Enrich Product Usages processes will augment the product usage records by adding data to the records from sources such as customer, product, or other reference data.
1.2.16.1.1	L4	Add Product Usage Data	Add data to the records from sources such as customer, product, or other reference data to augment the product usage records.

1.2.16.1.2	L4	Assign Product Usage Price	Assign a price to product usage without consideration of specific product or customer information. The assigned price may be used to enrich the product usage record.
1.2.16.2	L3	Guide and Assign Product Usages	The Guide Product Usages processes ensure that the event records used in the billing process relate to the correct customer billing account and products. A specific product usage record may be related to multiple customer billing accounts and subscribed products. // Distribution of product usage records to other processes may also occur.
1.2.16.2.1	L4	Assign Product Usages	Ensure that the Product Usages used in the billing process relate to the correct Product.
1.2.16.2.2	L4	Distribute Product Usage	Distribute billing event records to other processes.
1.2.16.2.3	L4	Guide Product Usages	Guide Product Usages process is in charge of identifying Product Usages based on Service Usages. / For example, for a Video on Demand where you can watch a video as many times as you want during 72 hours, several Service Usages might have been tracked (each time the user watches the video) and only one Product Usage will be identified for all Service Usages in the 72 hours after the first watch.
1.2.16.3	L3	Mediate Product Usages	The Mediate Product Usages process edits and reformats the data record to meet the needs of a recipient application.
1.2.16.3.1	L4	Edit Product Usages	Edit the data record for recipient applications.
1.2.16.3.2	L4	Reformat Product Usages	Reformat the data record for recipient applications.

1.2.16.4	L3	Report Product Usage Records	The purpose of the Report Product Usage Record processes is to generate reports on Product Usage records based on requests from other processes. / These processes produce reports that may identify abnormalities, which may be caused by fraudulent activity or related to customer complaints. / Investigation of problems related to these product usage records is also part of this process. / / These processes also support other processes such as customer review of product usages (pre-billing and post-billing).
1.2.16.4.1	L4	Generate Product Usage Report	Generate reports on product usage records based on requests from other processes.
1.2.16.4.2	L4	Investigate Product Usage Related Problem	Investigate problems related to product usage records.
1.2.16.4.3	L4	Support Product Usage Related Process	Support other processes such as customer review of product usages (pre-billing and post-billing).
1.2.17	L2	Product Rating & Rate Assignment	The purpose of Product Rating & Assignment is to rate a value (monetary or other) to Product Usage or a set of Product Usages and assign the result to a Product and a Billing Account. The charge may be either a credit or a debit and can be handled either online or offline. / / Online charging is performed in real-time, requiring an authorization component which may affect how the service is rendered and enables an operator to provide prepaid services to its customers. Whereas offline charging is performed after the service is rendered and is not required to be done in real-time and generally relates to subscription-based products.

1.2.17.1	L3	Perform Rating	Process responsible for calculating the value of a product usage or a set of product usages, before, during or after the rendering of the service, based on parameters of the request (type, quantity, etc.), parameters of the customer/subscriber (tariffs, price plans, accumulated usage, contracts, etc.) and other parameters (time-of-day, taxes, etc.). The same request may be rated differently for different subscribers based on their purchased offers or agreements.
1.2.17.2	L3	Aggregate Items For Rate Assignment	This process is responsible for accumulating contributing items, which can be quantities, values (monetary or other) or both. Aggregation can occur over time or can be initiated to gather a "snapshot" of the items at a point in time.
1.2.17.3	L3	Manage Customer Assignment Hierarchy	Customer hierarchies are commonly used for corporate customers, family plans or other types of affinity groups. This process manages the assignment relationships among subscribers, e.g. sharing, inheriting or restricting balances, price plans and discounts. Thereby assuring that a charge is added to or subtracted from the correct account balance.
1.2.17.4	L3	Provide Advice of Rate	The activity of Provide Advice of Rate (aka Advice of Charge) is responsible for providing advice on rates, in real-time or offline, an estimate or value of the rate for a specific usage request. The advice is usually based upon performing a full rating process for the request.

1.2.17.5	L3	Apply Rate Level Discounts	This process applies discounts to product prices at an individual product level. A discount may be expressed as a monetary amount or percentage and modifies a price for a product. When a discount is expressed as a percentage, the discounting process determines the discount calculated in relation to the price for the product. // The discount may be displayed as a separate entry on the bill or may be combined with the rate for the product to only show as one entry. // Discounts may be a one-time event or may have some duration (days, months, life of product, etc.). Discounts may apply to a specific customer or be generally available based on selection of products (for example - bundles). Discounting structures may involve tiers, tapers, or thresholds.
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2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
Product Usage	Product Usage ABE represents specification of usages enabled by Products and usages of products tracked.		<i>(no description available)</i>
Product Offering Instance	A ProductOfferingInstance represents the instantiation of a ProductOfferingSpecification for a specific Customer. It may carry ProductPrice and commitment choice.	Product Price	The Product Price ABE describes all the prices applied to Products.

2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
253	Customer Usage and Charges Report Access	/ Rating and Follow-up	/ Bill Calculation	Customer Usage and Charges Report Access provides an internet technology driven interface to the customer to access web-based reports for (historical) usage and charges directly for themselves.
55	Price and Discount Calculation	/ Rating and Follow up	/ Tariff Calculation and Rating	Price and Discount Calculation applies pricing and discounting rules and algorithms in the context of the assembled information concerning Products (i.e. instances of Product).
125	Charging Event Accumulation	/ Rating and Follow up	/ Tariff Calculation and Rating	Charging Event Accumulation function accumulates events that provide measurements that will be used in the charge calculation (e.g. used allowance).
126	Event Charge/Credit Calculation	/ Rating and Follow up	/ Tariff Calculation and Rating	Event Charge/Credit Calculation calculates event-level charges/credits (one time, recurring, and usage).
127	Calculated Charges/Credits Proration	/ Rating and Follow up	/ Tariff Calculation and Rating	Calculated Charges/Credits Proration provides proration of calculated charges/credits. The function handles partial rating of a period.

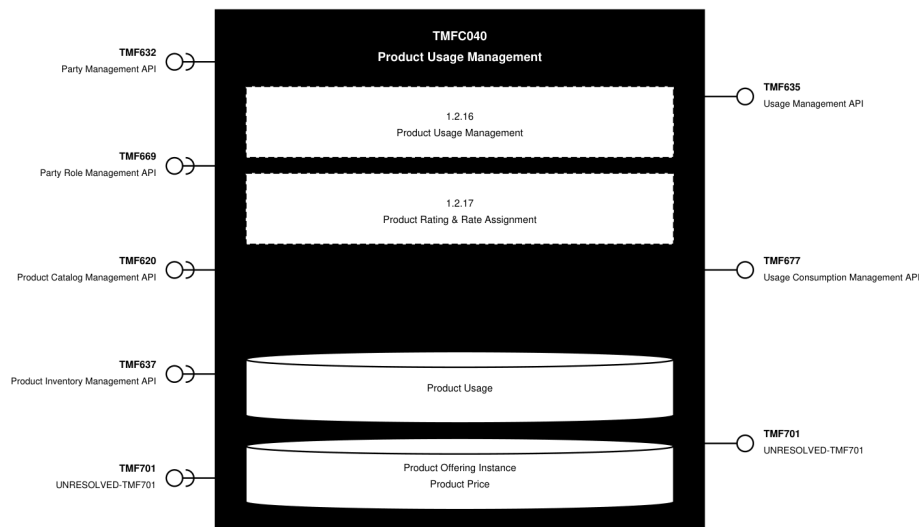
128	Late Arrival Usage Charges/Credits Recalculation	/ Rating and Follow up	/ Tariff Calculation and Rating	Late Arrival Usage Charges/Credits Recalculation provides recalculation of charges/credits based on information received later (e.g. from the Service Level Agreement function, delayed call detail record file arrival, delayed order arrival). Recalculation may be necessary: pre-billing (prior to Bill Calculation), during the Bill Calculation process, and/or post-billing.
186	Charging/Rating Recalculation	/ Rating and Follow up	/ Tariff Calculation and Rating	Charging/Rating Recalculation function recalculates the charges, when appropriate, across product, location, or customer, and considerations based on business rules.
300	Discounts Calculation	/ Rating and Follow up	/ Tariff Calculation and Rating	Discounts Calculation determines charge discounts based on pricing plan; including discounts on recurring, one time, and usage charges. Discounts may be applied at different levels such as cross product, cross location, or cross customer (all customers that are part of a given group plan – some affiliation). The discounts can be apportioned across multiple events.

67	Usage Summary and Details Presentation	Invoice Management	Invoicing	Usage Summary and Details Presentation presents usage summary and details (billed and non-billed) for a specific time period. **
85	Billing Event Processing Distribution	Invoice Management	Invoicing	Billing Event Processing Distribution function for correlation and distribution of usage data for processing of e.g. bill, customer and product. To supply relevant usage data to the Billing System.
86	Billing Event Processing Enrichment	Invoice Management	Invoicing	Billing Event Processing Enrichment with complementing data e.g. product or location information to supply relevant information to the Billing System.

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

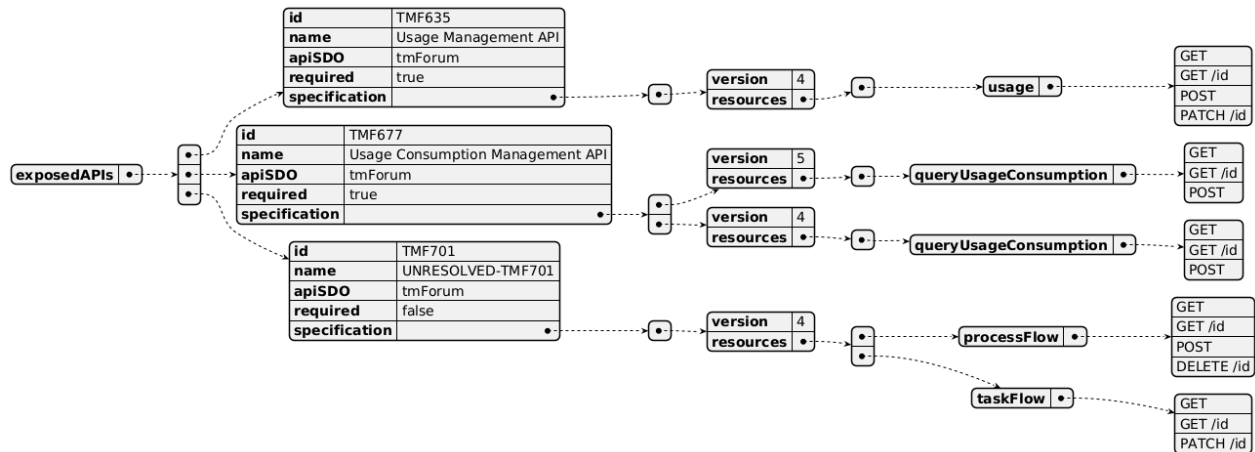
3.1. API Context Diagram



(SVG source: [TMFC040 API Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the *component-specification-documentation skill's scripts/render_api_context_svg.py*.)

3.2. Exposed APIs

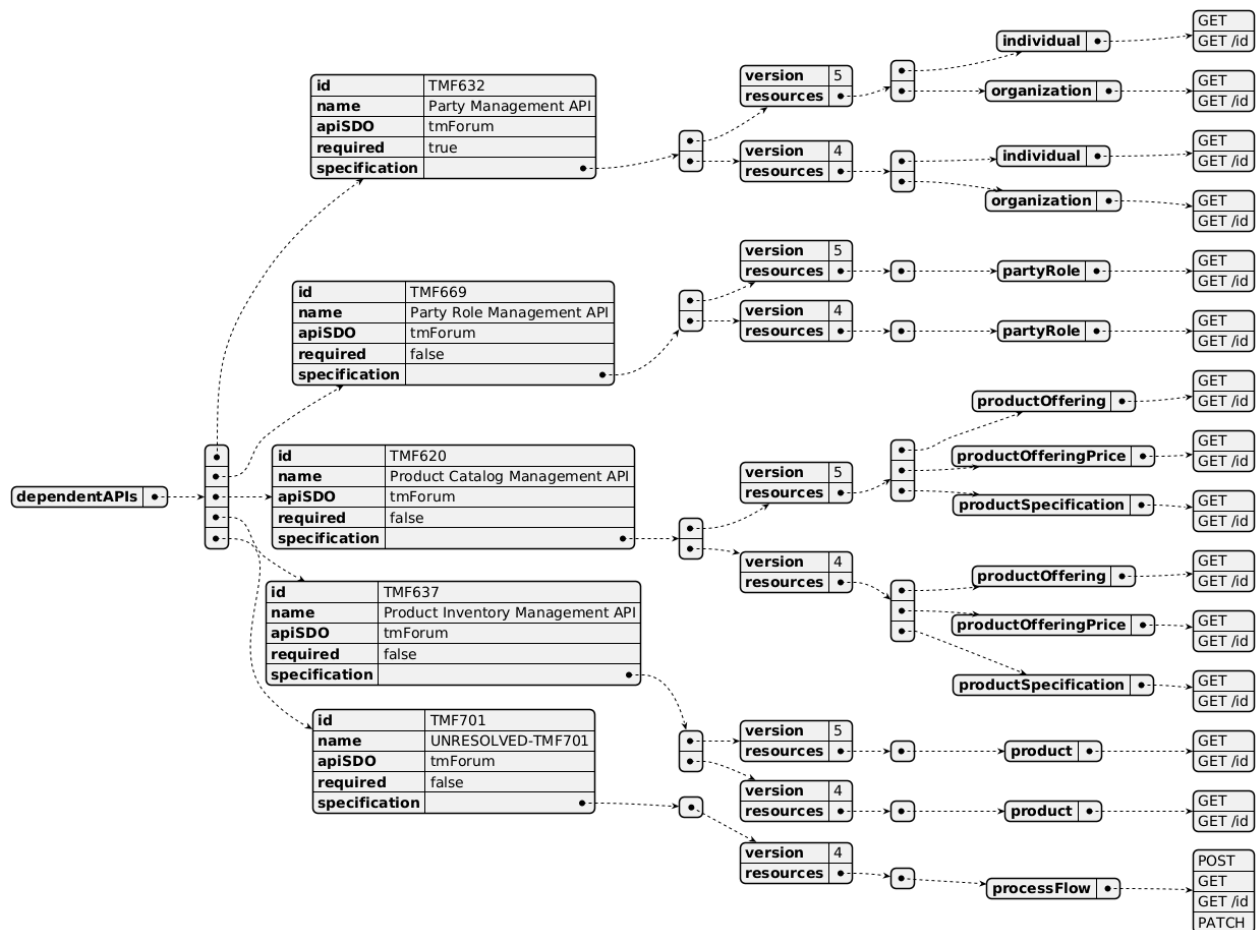
API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF635	Usage Management API	Mandatory	4	usage	GET, GET /id, POST, PATCH /id
TMF677	Usage Consumption Management API	Mandatory	5	queryUsageConsumption	GET, GET /id, POST
TMF677	Usage Consumption Management API	Mandatory	4	queryUsageConsumption	GET, GET /id, POST
TMF701	UNRESOLVED-TMF701	Optional	4	processFlow	GET, GET /id, POST, DELETE /id
TMF701	UNRESOLVED-TMF701	Optional	4	taskFlow	GET, GET /id, PATCH /id



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF632	Party Management API	5	Mandatory	individual	GET, GET /id
TMF632	Party Management API	5	Mandatory	organization	GET, GET /id
TMF632	Party Management API	4	Mandatory	individual	GET, GET /id
TMF632	Party Management API	4	Mandatory	organization	GET, GET /id
TMF669	Party Role Management API	5	Optional	partyRole	GET, GET /id
TMF669	Party Role Management API	4	Optional	partyRole	GET, GET /id
TMF620	Product Catalog Management API	5	Optional	productOffering	GET, GET /id
TMF620	Product Catalog Management API	5	Optional	productOfferingPrice	GET, GET /id
TMF620	Product Catalog Management API	5	Optional	productSpecification	GET, GET /id
TMF620	Product Catalog Management API	4	Optional	productOffering	GET, GET /id

TMF620	Product Catalog Management API	4	Optional	productOfferingPrice	GET, GET /id
TMF620	Product Catalog Management API	4	Optional	productSpecification	GET, GET /id
TMF637	Product Inventory Management API	5	Optional	product	GET, GET /id
TMF637	Product Inventory Management API	4	Optional	product	GET, GET /id
TMF701	UNRESOLVED-TMF701	4	Optional	processFlow	POST, GET, GET /id, PATCH

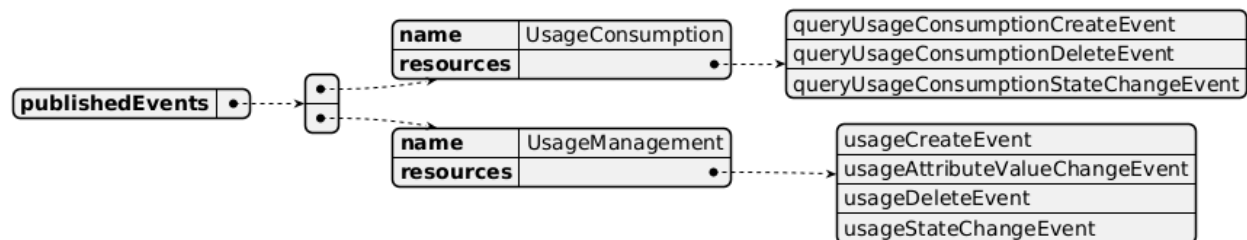


3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

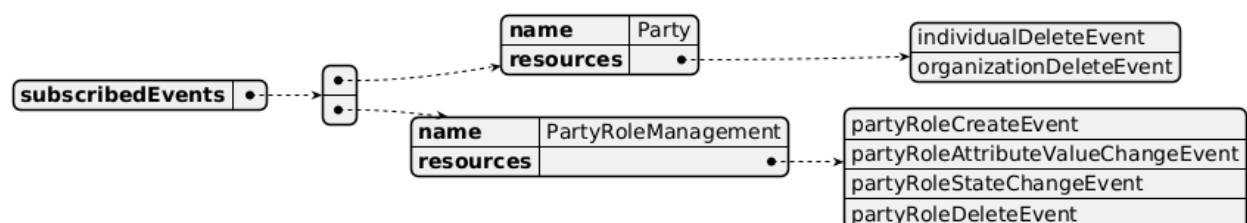
Published Events

API ID	API Name	Event Resources
TMF677	Usage Consumption Management API	queryUsageConsumptionCreateEvent queryUsageConsumptionDeleteEvent queryUsageConsumptionStateChangeEvent
TMF635	Usage Management API	usageCreateEvent usageAttributeValueChangeEvent usageDeleteEvent usageStateChangeEvent



Subscribed Events

API ID	API Name	Event Resources
TMF632	Party Management API	individualDeleteEvent organizationDeleteEvent
TMF669	Party Role Management API	partyRoleCreateEvent partyRoleAttributeValueChangeEvent partyRoleStateChangeEvent partyRoleDeleteEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.0
SID	v25.0
Functional Framework	v23.0

5.2. Jira References

(none currently listed)

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	27-Sep-2023	Gaetano Biancardi	Final edits prior to publication

1.1.0	08-Oct-2024	Elisabeth Andersson	Updated to the latest version of the Component Template. Added comment and Jira tickets related to Usage API evolution of v5. Clarified the scope of SID for this component in relation to ProductUsageSpecification. Changed requirements for TMF620 and TMF672 from Mandatory to Optional. Mapped to the latest Frameworks with the following changes: Functional Framework: name change of Aggregation level 1 and 2 (253, 55, 125, 126, 127, 128, 186, 300); added functions (65, 85, 86). Business Process Framework: Business Process Name change: 1.2.17.3
1.1.0	14-Nov-2024	Amaia White	Final edits prior to publication
1.1.1	20-Jan-2026	Rosie Wilson	Updates to description per agreed YAML updated on 03 Jun 2025. Final administrative updates prior to publication
1.2.0	25-May-2026	Hugo Vaughan	Gen 5 API updates
1.2.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	13-Feb-2024	Gaetano Biancardi	Initial release of document
Production	26-Apr-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	14-Nov-2024	Amaia White	Release of version 1.1.0
Production	27-Dec-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	20-Jan-2026	Rosie Wilson	Release of version 1.1.1
Production	20-Mar-2026	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	25-Jun-2026	Sannah Sibaya	Release of version 1.2.0

6.2. Acknowledgements

This document was prepared by the members of the TM Forum ODA Components & Canvas team.

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